# Simon Chen

347-322-4133 | shangminch@gmail.com | simon-chen.com | linkedin.com/in/shangmin-chen | github.com/shangmin-chen

## Education

**Boston University** — Boston, MA
*Bachelor of Arts in Computer Science* — *Graduated: Spring 2026*

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Databases, Machine Learning, Data Science.

## Experience

**Software Engineering Intern** — February 2026 – June 2026
*RESET Standard* — *Shanghai, China*

- Engineered asynchronous data pipelines with RabbitMQ, Redis, and Sidekiq to process long-running ingestion jobs in the background, decoupling external API calls from user-facing requests.
- Doubled the platform's supported environmental data categories (from 2 to 4) by independently implementing end-to-end water and waste data integration, developing Rails backend ingestion pipelines, PostgreSQL schemas, and client-facing React/Vite/Tailwind dashboards.
- Redesigned project-device mapping workflows in response to client requests, refactoring legacy forms to streamline onboarding of environmental monitoring devices.

**Lead Product Engineer** — December 2025 – Present
*EZ Esports* — *New York, NY*

- Designed a PostgreSQL schema on Supabase using Drizzle ORM, supporting 500+ matches across 20+ teams, and built RESTful endpoints for player statistics, standings, and historical data.
- Migrated an esports league platform from Google Sites to Next.js with a Supabase backend, replacing manual content updates with live scoring and automated season management.
- Cut page load time by 75% (from 3.2s to 0.8s) by implementing edge caching on Cloudflare, route-based code splitting, and asset optimization for the Next.js frontend.

## Projects

**Persephone** | *Python, Cython/C++17, Rust, Modal, FAISS* — May 2026

- Cut backtest and parameter-sweep iteration from ∼14 hours on laptop CPUs to ∼15 minutes by moving tick backtests and sweeps to Modal NVIDIA GPU workers with FAISS GPU search and CuPy-vectorized math.
- Replaced KDE inference with a GBM theo layer, cutting model query latency from ∼1.5ms to ∼20$\mu$s per query (∼75x) behind an unchanged live serving interface.
- Sped up 155K-row neighbor search from 758$\mu$s to 154$\mu$s p50 by caching FAISS IVF indexes and pinning FAISS threads to avoid OpenMP oversubscription.
- Rewrote 9 live hot-path kernels (order-book maintenance, order-book imbalance, trade intensity, event aggregation) as native Cython extensions, cutting Kalshi book maintenance from 21.8ms to 195$\mu$s p50 (112x).
- Designed a lock-free SPSC event ring in Cython/C++17 with cache-line-aligned atomic cursors and packed uint64 records, benchmarked at 584ns per enqueue and 703ns per event drained.
- Prevented bad-price quoting with fail-closed gates that block stale, crossed, or untrusted market data before quote generation, computing fused market metrics in 357$\mu$s p50 across 8 venues.

**Whisperrr** | *Java Spring Boot, Python FastAPI, React, PostgreSQL / Tailwind CSS* — September 2025

- Built full-stack transcription platform processing audio at 32x realtime speed, supporting 99+ languages across multiple audio formats with real-time results
- Designed three-tier stateless architecture with HTTP connection pooling to reduce request latency by 70%, improving reliability for files over 25MB from 60% to 100% success rate
- Containerized microservices with Docker Compose health checks and implemented segment-level timestamp generation for subtitle creation

## Technical Skills

**Languages**: Python, C++17, Cython, Rust, SQL, TypeScript, JavaScript, Java, Ruby
**Frameworks & Libraries**: FAISS, CuPy, FastAPI, React, Next.js, Ruby on Rails, Django, Spring Boot, Tailwind CSS
**Databases & Infrastructure**: Modal, OpenMP, Docker, PostgreSQL, Redis, RabbitMQ, Sidekiq, Cloudflare, Nginx